

# Oliver Rupp

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## Education

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### University of Notre Dame

B.S. in Physics, Honors Track; B.S. in Mathematics

*Concentrations:* Astrophysics and Advanced Physics

*Notre Dame, IN*

Expected May 2027

GPA: 4.0

*Relevant Coursework:* Quantum Mechanics I–II, Relativity: Special and General, Electromagnetic Waves, Modern Observational Techniques, Advanced Mathematical Methods, Complex Variables, Abstract Algebra

## Research Experience

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### University of Notre Dame

*Research Assistant — Theoretical & Computational Astrophysics*

*Advisor: Dr. Grant Mathews*

*Notre Dame, IN*

Spring 2024–Present

- Simulated core-collapse supernovae with the GR1D hydrodynamics code to investigate the effects of hadron–quark phase transitions on supernova evolution and explodability.
- Developed Python tools to convert hybrid hadron–quark equation-of-state (EOS) tables from CompOSE into the format required by GR1D, including interpolation, physical definitions, and unit conversions.
- Identified and resolved numerical instabilities in GR1D during the hadron–quark phase transition through dynamic Courant-factor reduction and implicit variable updates.
- Analyzed explodability and shock dynamics across EOS models; results presented at APS DNP CEU and incorporated into a manuscript in preparation.
- Developed a Python-based neutron-star oscillation model using the TOV equations and a Sturm–Liouville eigenvalue problem to investigate radial oscillations in simulations.

### University of Utah

*NSF REU Researcher — Theoretical Astrophysics*

*Advisor: Dr. Tanmoy Laskar*

*Salt Lake City, UT*

Summer 2026

- Benchmarked parameter recovery across three gamma-ray burst afterglow modeling codes: GAMMA, PyFRS, and VegasAfterglow.
- Developed a model-agnostic Python MCMC program to automate parameter sampling and analyze fitting degeneracies across afterglow models.
- Presented results at the Utah Summer Research Symposium.

### University of Notre Dame

*Research Assistant — Observational Astronomy*

*Advisor: Dr. Lauren Weiss*

*Notre Dame, IN*

Fall 2026–Present

- Validate or demote transiting exoplanet candidates identified by NASA TESS, NASA Roman, and ESA PLATO using the Sarah L. Krizmanich Telescope.
- Analyze telescope observations using differential aperture photometry and MaxIm DL and AstroImage.

### University of Notre Dame

*Modern Observational Techniques Course - Observational Astronomy*

*Advisor: Dr. Jeffrey Chilcote*

*Notre Dame, IN*

Fall 2025

- Observed M52 and NGC 884 using a 0.8 m telescope and determined the distance to NGC 884 through differential photometry and main-sequence matching.
- Measured the period of the SX Phoenicis variable CY Aquarii.

## Selected Manuscript

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**Rupp, O.**, Mathews, G., & Boccioli, L. *Dynamics and Explodability of Core Collapse Supernovae with a Transition to Quark Matter*. Manuscript in preparation, 2026.

## Presentations

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- **Comparing Parameter Recovery Across GRB Afterglow Models** — Utah Summer Research Symposium, 2026.
- **Explodability of CCSNe with Transition to Quark Matter** — APS DNP CEU, 2025; Notre Dame Undergraduate Research Fair, 2025.
- **Mass Constraints on Primordial Black Holes** — Notre Dame Undergraduate Research Fair, 2024.

## Technical Skills

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**Programming:** Python (NumPy, SciPy, Matplotlib, AstroPy, emcee, h5py), Fortran, C++, Mathematica

**Computing:** Linux-based HPC clusters, Git/GitHub, LaTeX

**Observational Astronomy:** 0.8m telescope operation, MaxIm DL photometry software

## Teaching & Leadership

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### Society of Physics Students

*University of Notre Dame*

*Co-President*

Fall 2026–Present

- Coordinate chapter meetings, delegate responsibilities across the executive board, and organize events including game nights, physics formal, and other community-building activities.
- Created a new undergraduate research presentation night to introduce underclassmen to research opportunities and encourage early involvement in physics research.
- Collaborate with the graduate outreach community on public outreach events connecting Notre Dame physics students with the broader community.

### Cosmos Astronomy Club

*University of Notre Dame*

*Co-President*

Fall 2025–Present

- Organize and lead stargazing events featuring planets, stellar clusters, and other astronomical objects for the Notre Dame community.
- Volunteer with astronomy outreach events on campus and in the local community.

### University of Notre Dame

*Notre Dame, IN*

*Astronomy Teaching Assistant*

Fall 2024–Present

- Lead nighttime telescope observations and teach students to identify and interpret astronomical objects using computerized telescopes.

### Society of Physics Students

*University of Notre Dame*

*Secretary*

Fall 2025–Spring 2026

- Organized elections, outreach, mentoring, and social activities; improved connections between undergraduate and graduate students and led a graduate-school admissions workshop.

## Awards & Honors

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- APS DNP CEU Selected Participant, 2025
- Notre Dame College of Science Summer Undergraduate Research Fellowship, 2025
- Dean's List, Fall 2023–Present